

Center, spread, and distributions

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HES 3000-002, Sport Performance Analytics

levels of granularity

Player Batting Stats - All Splits

NAME	GP	AB	R	H	2B	3B	HR	RBI	TB	BB	SO	SB	AVG	OBP	SLG	OPS	WAR
Freddie Freeman 1B	150	568	74	165	32	1	16	70	247	66	113	7	.290	.365	.435	.800	2.7
Kyle Tucker RF	150	524	88	124	23	4	17	70	206	77	102	12	.237	.334	.393	.728	1.1
Shohei Ohtani DH	131	506	89	140	25	4	30	80	263	81	146	11	.277	.381	.520	.900	3.4
Andy Pages CF	130	499	73	137	26	2	21	80	230	46	108	12	.275	.342	.461	.803	5.8
Max Muncy 3B	146	479	80	121	26	1	29	76	236	63	138	3	.253	.347	.493	.840	4.2
Mookie Betts SS	119	464	69	119	24	0	22	68	209	45	55	2	.256	.324	.450	.775	3.7
Teoscar Hernandez LF	121	429	65	116	17	2	15	63	182	42	129	5	.270	.338	.424	.762	0.8
Tommy Edman 2B	81	280	43	68	11	2	8	41	107	21	65	6	.243	.326	.382	.708	1.7
Alex Freeland 2B	89	231	32	54	9	0	4	21	75	32	74	4	.234	.327	.325	.652	1.2
Will Smith C	71	228	34	57	9	0	10	34	96	34	43	0	.250	.349	.421	.770	1.5
Dalton Rushing C	78	221	37	55	12	0	12	36	103	23	73	0	.249	.332	.466	.798	1.4
Miguel Rojas 2B	102	219	22	59	16	0	4	22	87	14	26	1	.269	.310	.397	.707	0.7
Alex Call LF	86	160	24	37	8	0	1	19	48	23	35	1	.231	.349	.300	.649	0.2
Hyeseong Kim SS	43	116	16	30	3	1	1	11	38	12	31	5	.259	.323	.328	.651	0.5
Enrique Hernandez 3B	38	81	11	18	4	0	3	11	31	6	27	0	.222	.276	.383	.659	0.0
Hunter Feduccia C *	30	67	6	15	2	0	1	7	20	13	15	0	.224	.350	.299	.649	0.1
Santiago Espinal 3B	36	56	5	15	3	0	1	7	21	1	8	0	.268	.276	.375	.651	0.0
Ryan Ward LF	20	55	8	12	4	0	3	12	25	5	17	0	.218	.283	.455	.738	0.2
Josue De Paula DH	11	31	7	10	3	1	1	7	18	11	5	1	.323	.500	.581	1.081	0.5
Eliezer Alfonzo C	13	28	3	5	2	0	0	4	7	0	3	0	.179	.179	.250	.429	-0.2
Ben Rortvedt C	16	27	0	5	0	0	0	1	5	0	9	0	.185	.185	.185	.370	-0.1
Chuckie Robinson C	8	23	1	2	0	0	0	1	2	0	5	0	.087	.087	.087	.174	-0.4
Alek Thomas CF *	7	19	1	4	1	0	0	3	5	1	4	0	.211	.250	.263	.513	-0.1
Total	158	5311	788	1368	260	18	199	744	2261	616	1231	70	.258	.339	.426	.765	

HITS		H
1	 Milwaukee Brewers	1,381
2	 Pittsburgh Pirates	1,374
3	 Los Angeles Dodgers	1,368
4	 Chicago Cubs	1,365
4	 Tampa Bay Rays	1,365

RUNS BATTED IN		RBI
1	 Chicago Cubs	793
2	 Milwaukee Brewers	766
3	 Washington Nationals	765
4	 Los Angeles Dodgers	744
5	 Pittsburgh Pirates	717

BATTING AVERAGE		AVG
1	 Los Angeles Dodgers	.258
1	 Tampa Bay Rays	.258
3	 Milwaukee Brewers	.257
4	 Pittsburgh Pirates	.254
5	 Colorado Rockies	.253

Summarizing your data

- A single number can answer different questions:

Question	Method	Example
How much happened?	Sum	Total points scored
How much per game?	Rate	Points per game
What is typical?	Center	Mean or median points in a game
How much does it vary?	Spread	Range or standard deviation

WHAT IS “TYPICAL”?

Measures of central tendency

- A method of describing a data set with a single figure
- Main methods:
 - Mean
 - Median
 - Mode
- Most appropriate method depends on the data and question
 - Scale of measurement
 - Outliers and data distribution

Mean, Median and Mode

- Mean: arithmetic mean
 - Determined by the sum of the values in the distribution divided by the number of values.
 - Misleading if outliers or asymmetries are present
- Median: the middle value
 - the data point below which half of the values lie
 - Ignores magnitude, focused on order
- Mode: most frequent value
 - Rarely used for interval or ratio data

Mean, Median and Mode

- Mean: arithmetic mean
 - =AVERAGE(range)
- Median: the middle value
 - =MEDIAN(range)
- Mode: most frequent value
 - =MODE.SNGL(range)

How much does it vary?

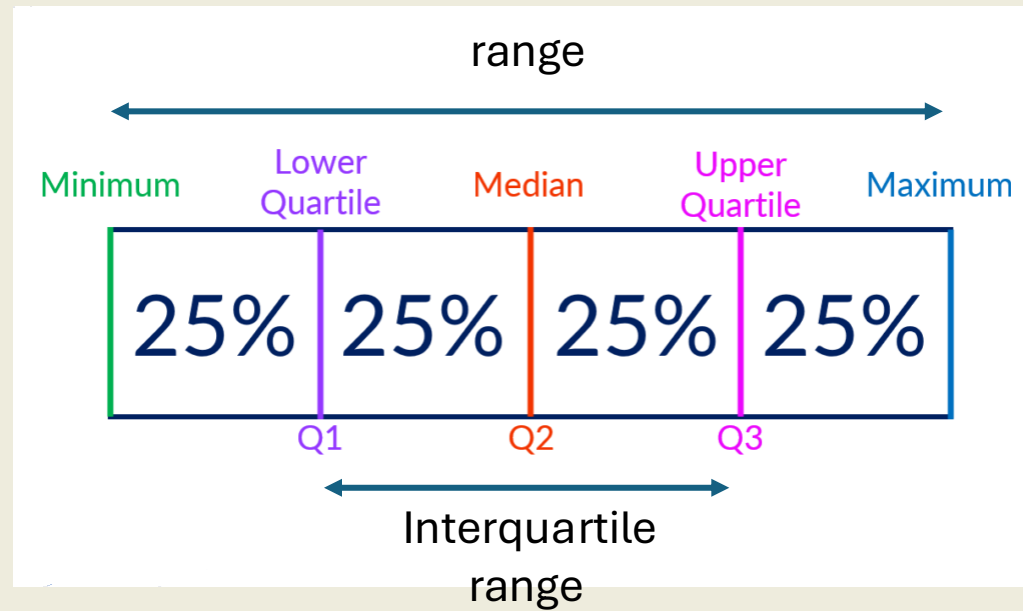
Measures of variability

- The degree of spread or distribution of the data.
- Can be used to show consistency or meaningful differences.
- Methods:
 - Range
 - IQR
 - Variance
 - Standard Deviation

Range

- Range: distance between the smallest and largest values.
 - Sensitive to outliers
- IQR: spread of the middle 50% of the data.
 - Less sensitive to outliers
- Standard Deviation: square root of variance
 - Typical deviation from the mean
 - Same units as the data

Range and IQR

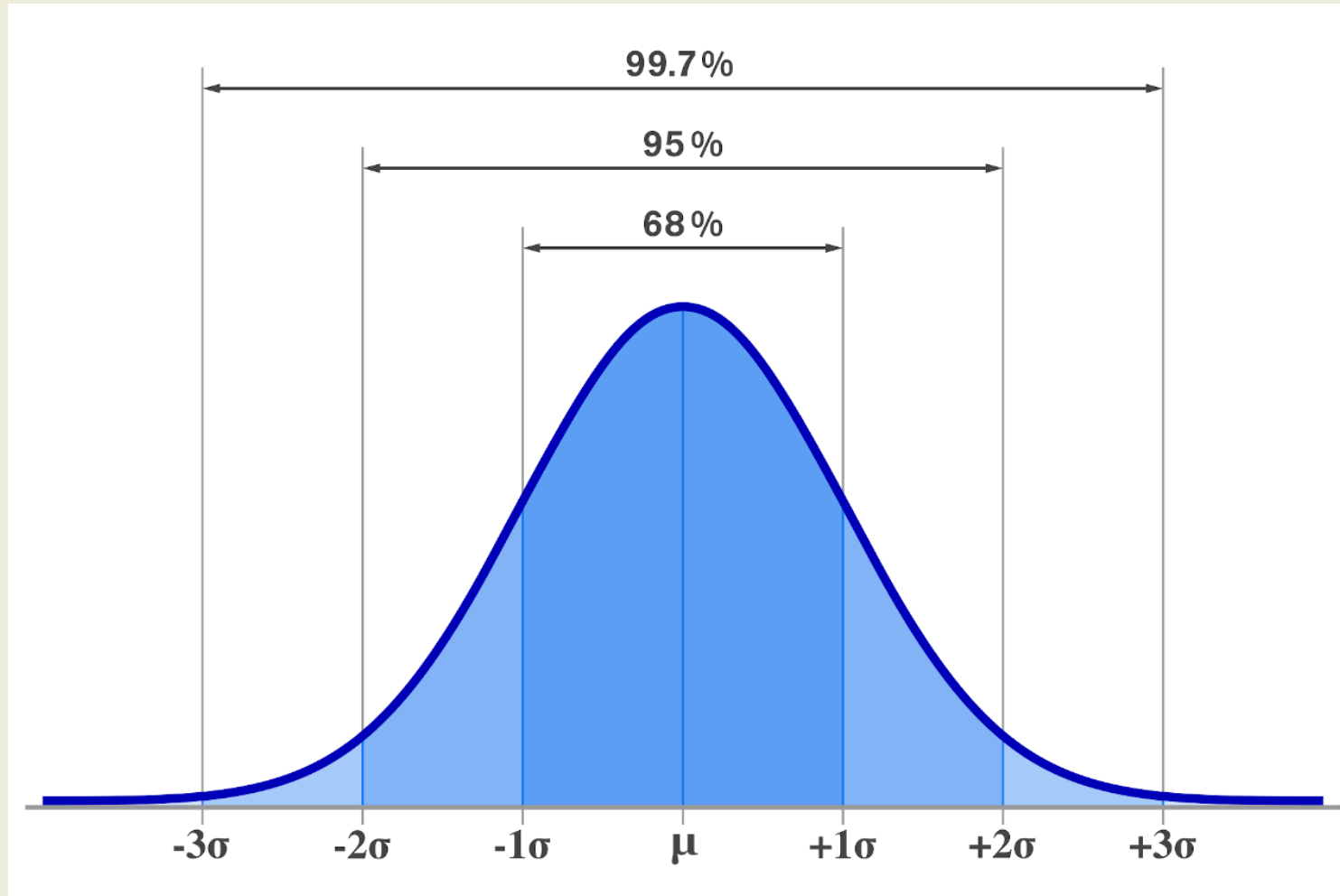


Standard Deviation

- Range: distance between the smallest and largest values.
 - Sensitive to outliers
- IQR: spread of the middle 50% of the data.
 - Removes outliers
- Standard Deviation: Typical deviation from the mean
 - square root of variance
 - Same units as the data
 - =STDEV.S(range)

Standard Deviation

Deviation

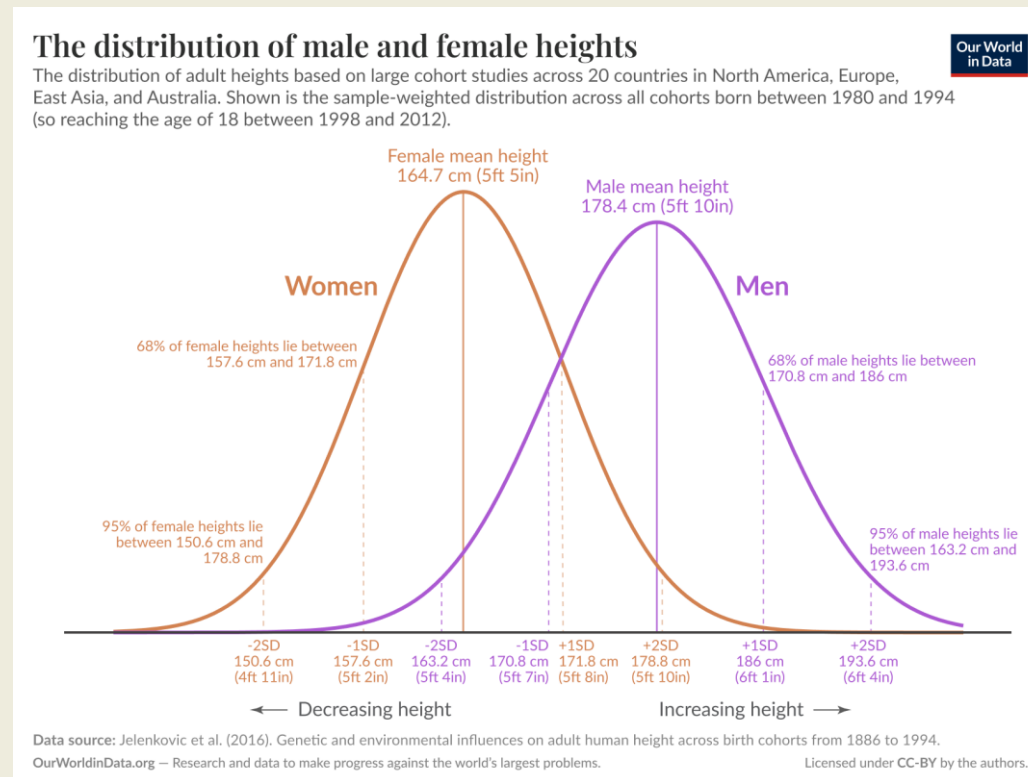


mean

Distribution in data

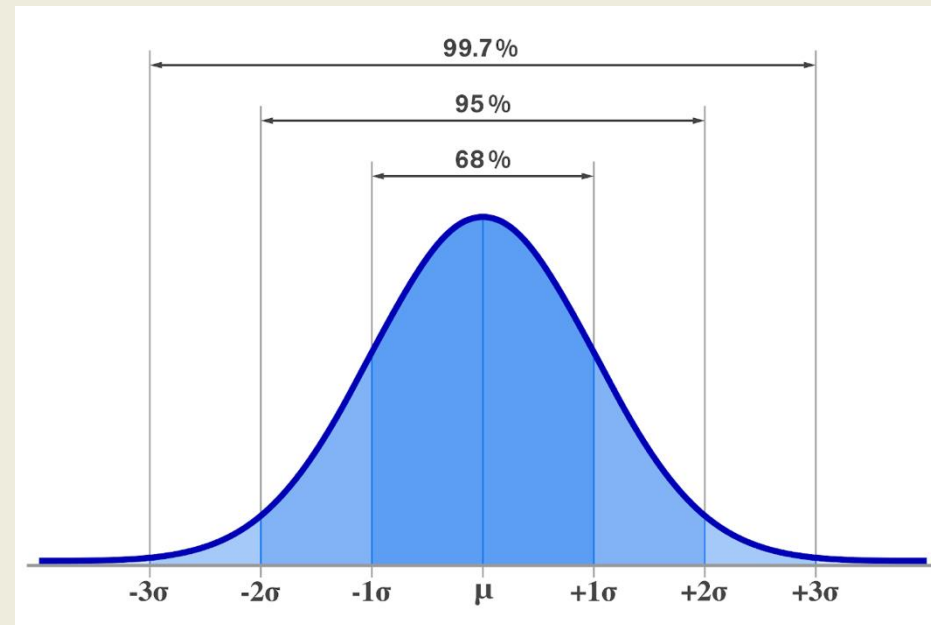
Understanding Distributions

- The Normal Distribution (Bell Curve):
 - Many biological and physical traits roughly approximate a normal distribution



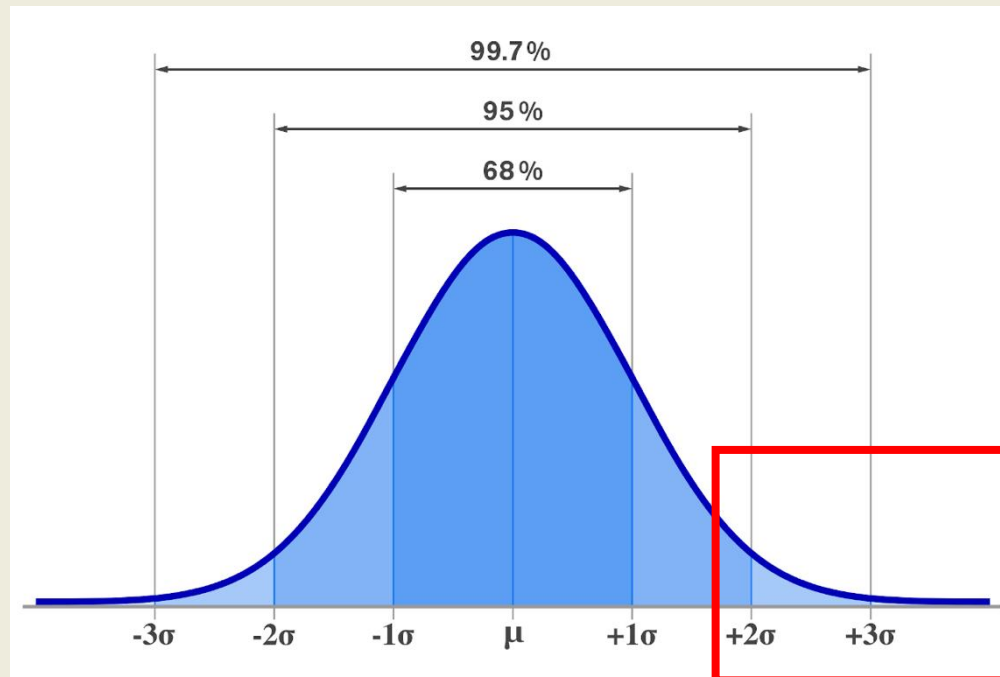
Understanding Distributions

- The Normal Distribution (Bell Curve):
 - Many biological and physical traits roughly approximate a normal distribution
 - Defined by mean and standard deviation (symmetric).



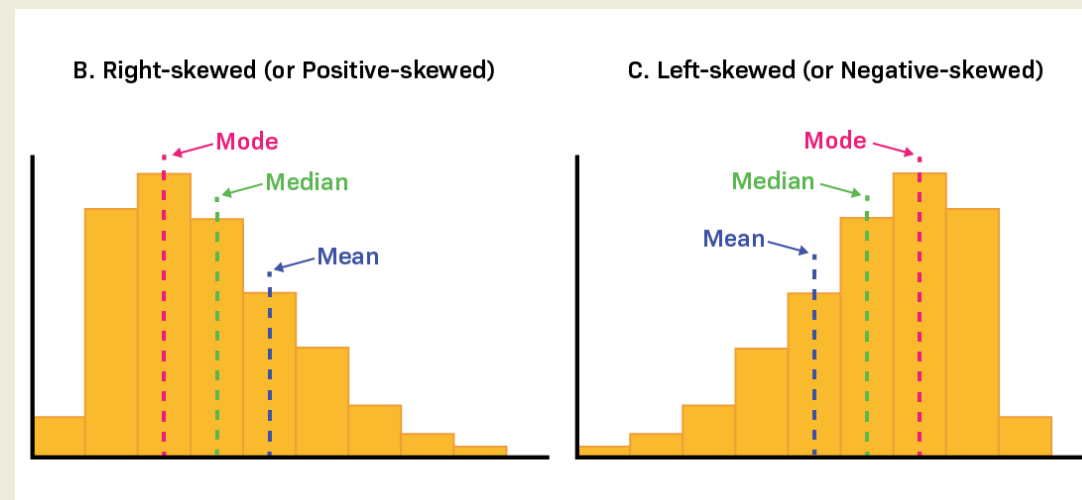
Distribution in Sports

- Data in sports often not-normal
 - Pooling athletes can create skew because athletes have different roles and opportunities.
 - Measurements within one athlete may be more symmetric



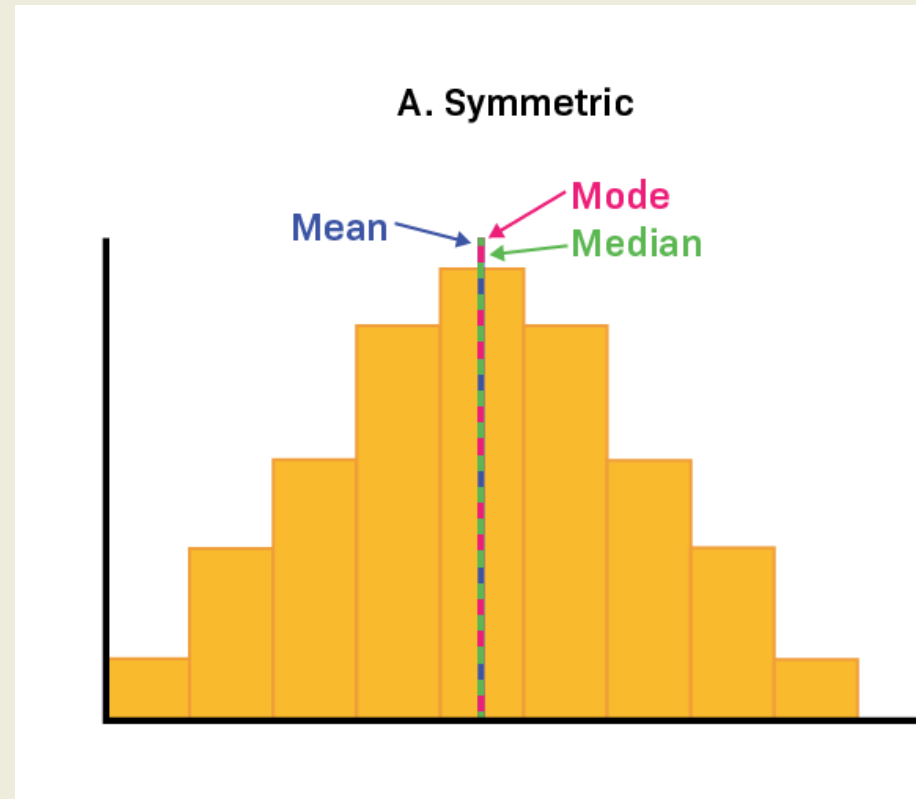
Skew in Data

- Right-Skewed (Positive Skew):
 - Most players have low counts, but a few superstars skew the tail high.
- Left-Skewed (Negative Skew):
 - Values clustered near the maximum ceiling, with a few poor performances trailing off.

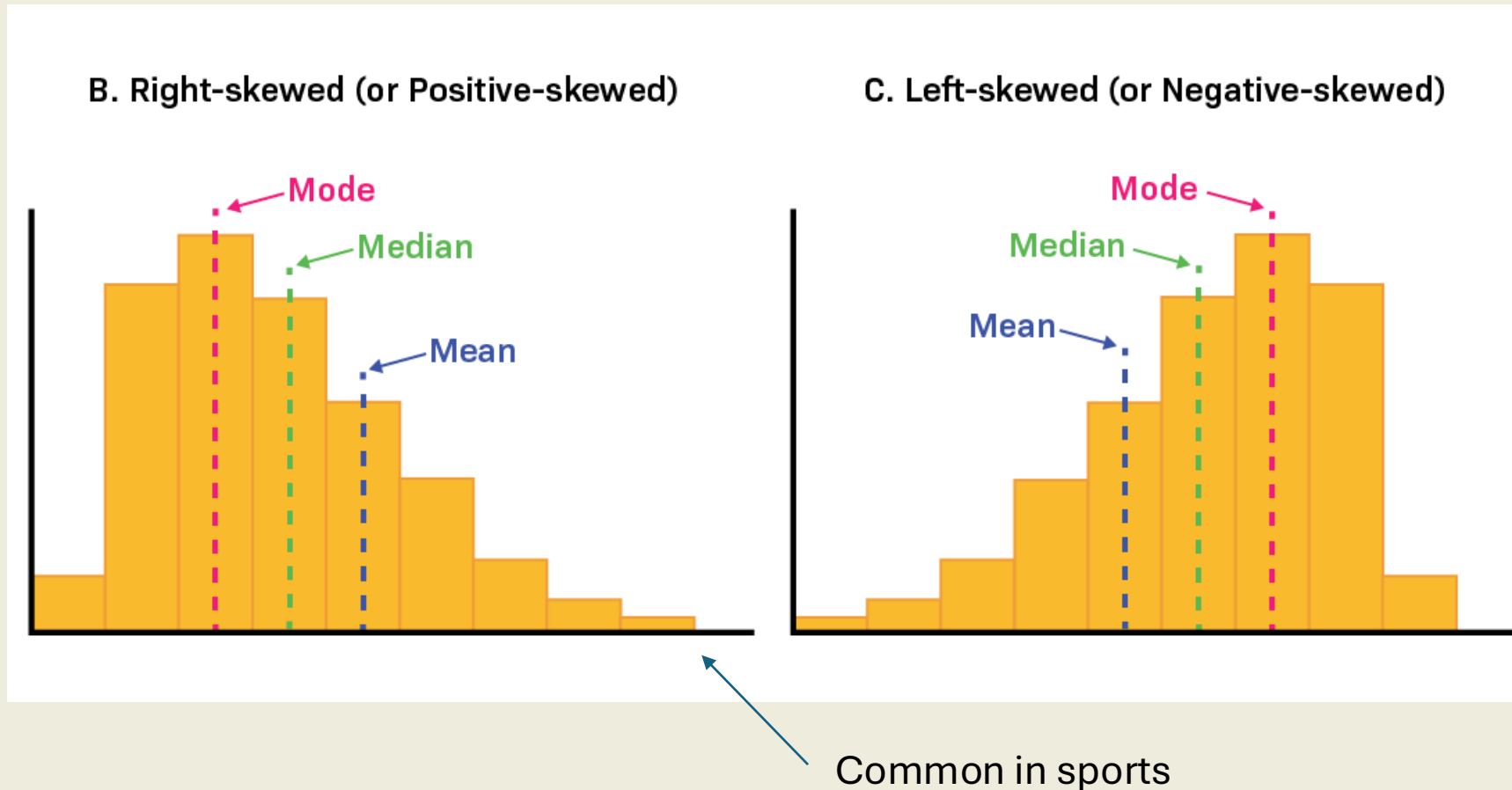


Measuring the center in non-normal data

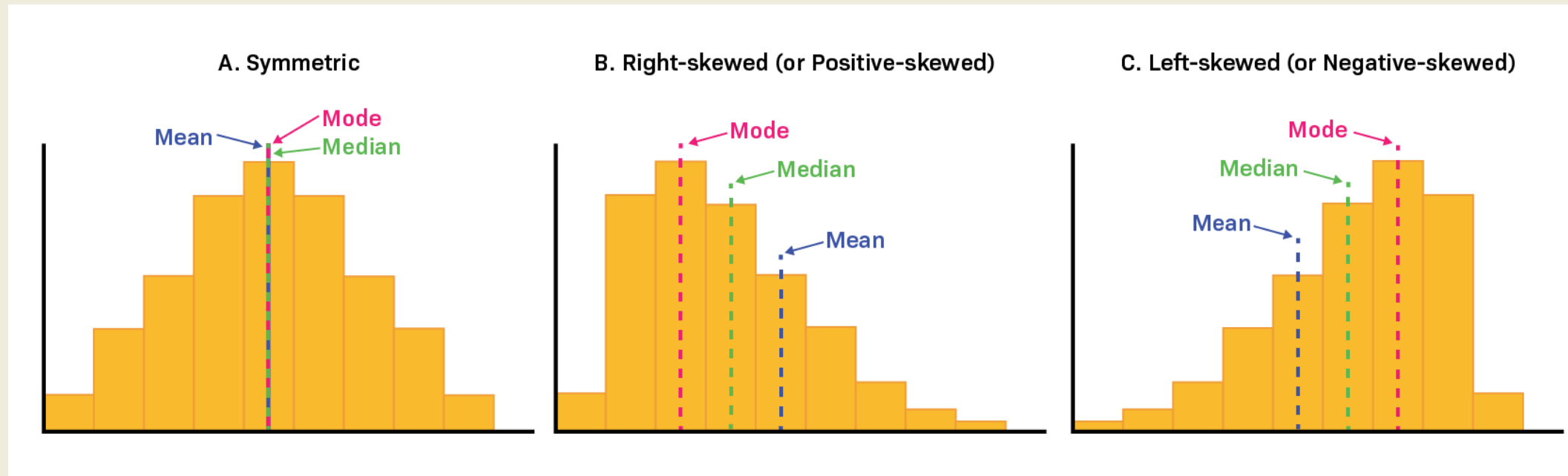
The center in symmetric data



The center in asymmetric data



Does distribution change appropriate method?



Would you / should you change the averaging method based on distribution?

Grouped summary

Grouped summary statistics

Why Segment Your Data?

- Looking at team / league average often masks the truth because different positions have entirely different physical and tactical requirements.
- Example:
 - Grouping all positions may make total high-speed distance look "average."
 - Segmenting by position may reveal that certain groups log massive high-speed workloads.

Grouping can change the conclusion

Who is the better shooter?

	2pt made	2pt att	3pt made	3pt att
Player A	12	16	1	4
Player B	4	4	8	16

	2pt%	3pt%
Player A	75%	25%
Player B	100%	50%

Different grouping can flip the conclusion

Who is the better shooter?

- $FG\% = (2\text{pt made} + 3\text{pts made}) \div (2\text{pt attempted} + 3\text{pts attempted})$

	2pt%	3pt%	FG%
Player A	75%	25%	65%
Player B	100%	50%	60%

Player B

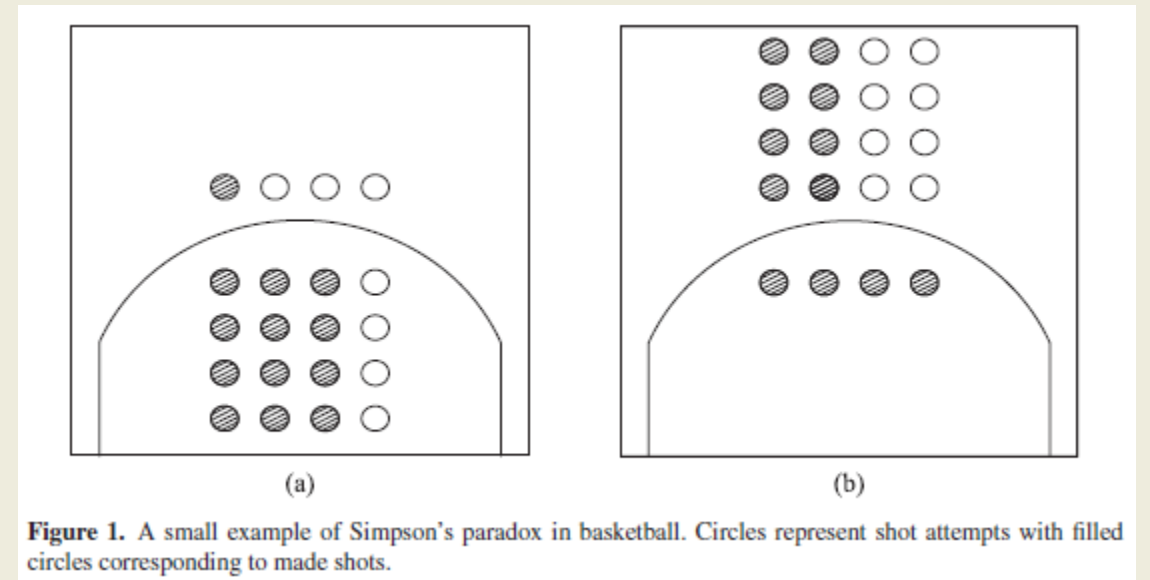
Player A

How can this be??

Simpson's paradox

A pattern that appears within each group may reverse when the groups are combined, because the groups contribute different amounts to the overall result.

	2pt%	3pt%	FG%
Player A	75%	25%	65%
Player B	100%	50%	60%



Visualizing Distributions and Groups

- Numbers tell part of the story, but visual shapes reveal the **full picture**.
- **Bar graphs, scatter plots** and **line graphs** can show averages and averages over time
- **Histograms, density plots**, and **boxplots** are useful for spotting positional differences and skewness.

Summary statistics cuts out nuance

Steph Curry's 2021-2022 NBA Playoff shot locations

- Mean, standard deviation and correlation of shot locations.

mean of x	0.73977
mean of y	14.71046
standard deviation of x	11.26635
standard deviation of y	10.42091
correlation of x and y	-0.00989

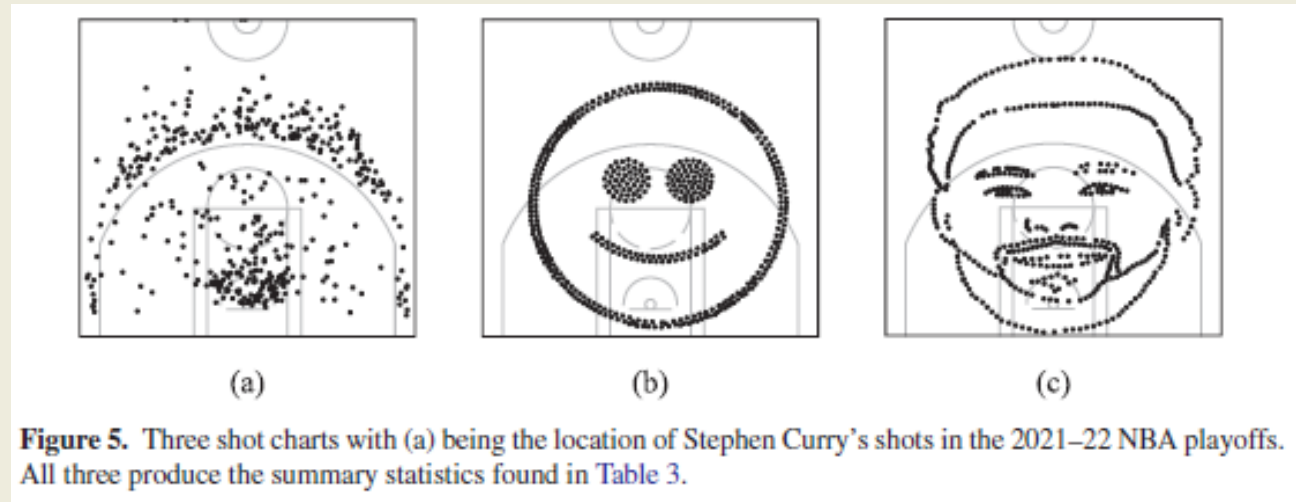
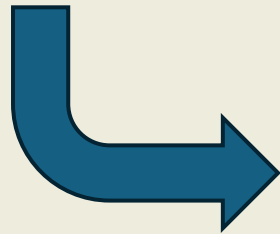
Table 3. Summary statistics for Stephen Curry's shot locations (in feet) in the 2021–22 NBA playoffs.

Visuals show the details

Steph Curry's 2021-2022 NBA Playoff shot locations

- Simple summary statistics can obscure actual data distribution (shot location).

mean of x	0.73977
mean of y	14.71046
standard deviation of x	11.26635
standard deviation of y	10.42091
correlation of x and y	-0.00989



Anscombe's quartet

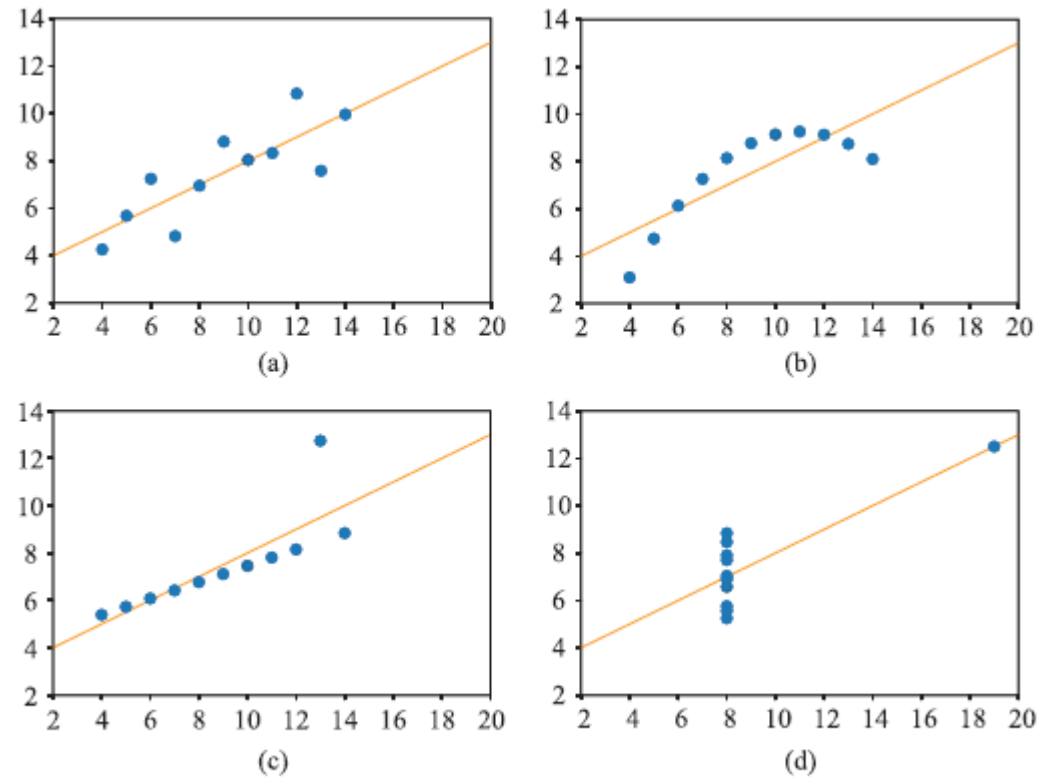


Figure 4. Anscombe's quartet have similar summary statistics but exhibit significant differences when plotted.

In-Class Case Study: Describing Your Project Data

In-Class Case Study

- Before conducting analysis, it is good to find out what your data actually looks like. In this activity, you will describe one important numeric variable using measures of center, spread, and distribution.

In-Class Case Study

Part 1: Choose what to describe

- Identify what **one row** of your dataset represents: a player, a player-game, a team-game, a training session, or something else.
- Choose one numeric variable that matters.
- Record the variable's units and the number of observations with a usable value.

If you do not have data, use a provided class dataset and complete the same steps. Note which dataset you used.

In-Class Case Study

Part 2: Summarize the variable

- Calculate:
 - Mean and median
 - Minimum and maximum
 - Range
 - Standard deviation
- Make one plot that shows the distribution

In-Class Case Study

Part 3: Interpret what you found

- Write three or four sentences that address these questions:
 1. What would you tell someone is a **typical** value? Would you use the mean or median, and why?
 2. How much do observations vary? Use at least one measure of spread in your answer.
 3. What does the distribution show that the summary statistics alone might hide? Consider unusual values, skew, gaps, or clusters.

In-Class Case Study

Bonus:

- Compare the variable across a relevant group, such as position, team, season, or competition level.
 1. Report the number of observations in each group and compare both center and spread.
 2. Describe what you observe without assuming the group difference has a particular cause.